PREOPERATIVE DIAGNOSIS: Obstructive 8 mm distal left ureteral calculus with severe hydroureteronephrosis and intractable left flank pain.

POSTOPERATIVE DIAGNOSIS: Impacted 8 mm distal left ureteral calculus successfully fragmented; left double-J stent placed.

PROCEDURE: Cystourethroscopy, retrograde pyelogram, left semi-rigid ureteroscopy, holmium laser lithotripsy, stone basket extraction, and insertion of a 6-French x 26 cm double-J ureteral stent.

PROCEDURE NARRATIVE: Under general laryngeal mask anesthesia, the patient was positioned in the dorsal lithotomy position. Rigid cystoscopy was performed using a 21-French cystoscope. The left ureteral orifice was identified, exhibiting edema. A 0.038-inch sensor PTFE-coated guidewire was passed through the left ureteral orifice into the renal pelvis under fluoroscopic guidance. Retrograde pyelogram was executed with 50% diluted Isovue-300, confirming moderate left hydronephrosis with an obstructive filling defect at the level of the left sacroiliac joint. A second safety guidewire was introduced.

Over the working guidewire, a 6.0/7.5-French Wolf semi-rigid ureteroscope was gently introduced into the left ureteral orifice. The ureter was inspected under gravity irrigation. At 5 cm from the ureteral orifice, a yellow, irregular, calcium oxalate monohydrate-appearing stone measuring approximately 8 mm was visualized, firmly impacted in the mucosa. A 272-micron holmium:YAG laser fiber was advanced through the ureteroscope. Laser lithotripsy was initiated at settings of 0.8 Joules and 10 Hz (8 Watts) utilizing a dusting technique to fragment the periphery of the stone, followed by fragmentation settings of 1.2 Joules and 8 Hz to break the core into fragments <1.5 mm. Multiple larger residual fragments were extracted using a 1.9-French zero-tip nitinol basket.

Following complete stone clearance, retrograde pyeloscopy demonstrated complete fragment resolution and absence of ureteral mucosal avulsion or perforation. Under fluoroscopic guidance, a 6-French x 26 cm Cook Universa double-J ureteral stent was advanced over the guidewire; proximal coil in the renal pelvis and distal coil in the urinary bladder were confirmed. A 16-French Foley catheter was placed to gravity drainage.